Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate the potential energy gained by the car.

- Ep = mgh = 650 × 9.8 × 12
- Ep = 76 440 J (76 kJ)

3 Quick check

- a) Mass and velocity
- b) Watt
- c) Heat and sound
- d) 27 000

4 Key terms

- 1 – F
- 2 – C
- 3 – B
- 4 – D
- 5 – E
- 6 – A

5 Calculate the kinetic energy.

- Ek = ½mv² = 0.5 × 110 × 7²
- Ek = 2695 J (2.7 kJ)

6 Calculate the energy used in 3 minutes.

- t = 180 s; E = Pt = 2200 × 180
- E = 396 000 J (400 kJ)

No marks for: Using 3 instead of 180 seconds

7 Calculate the heat energy needed.

- Eh = cmΔT = 4180 × 1.5 × 85
- Eh = 532 950 J (530 kJ)